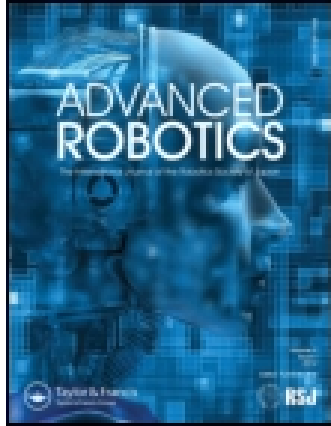


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THROO: a Tracked Hybrid Rover to Overpass Obstacles

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FULL PAPER

THROO: a Tracked Hybrid Rover to Overpass Obstacles

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In this paper, the design, simulation and experimental tests are presented for THROO: a Tracked Hybrid Rover, which has been developed to Overpass Obstacles. The proposed mobile robot has 3-DOFs and it is capable of straight motion, turning ability and two operations, namely rover-like motion with tracks on flat terrain and walking-like motion with track and legs to overpass obstacles to move on uneven terrain. The leg mechanism is composed of a four-bar linkage, which has been synthesized according to the desired features. In particular, the Burmester problem, which aims at finding the geometric parameters of a four-bar linkage required for a prescribed set of finitely separated poses are solved for the case under study. Dynamic simulations have been carried out and a prototype has been built. The proposed results show the hybrid rover ability to overpass obstacles, for which size is comparable or greater than the track high.

Keywords: mobile robots; hybrid robots; kinematic synthesis; simulation; experimental tests

1. Introduction

Mobile robots represent an interesting field of research due to their great potential features. They can be used for a wide range of applications including inspection,[1] service,[2] defence, manufacturing, cleaning, remote exploration,[3] rescue [4] and entertainment. Mobile robots can be classified according to the type of locomotion they perform as wheeled, tracked, legged or hybrid robots.[5] Wheeled and tracked systems can perform fast and have robust motion on flat terrain, but they are ineffective in overpass obstacles that are greater, or even with similar size, than the radius (dimension) of the wheel (track). Theoretically, legged robots can go where wheeled robots cannot, so they have a potentiality that justifies the extra effort required to control their locomotion.[6] It is obvious by reported works that legged robots have favourable characteristics for obtaining flexible and robust motion, but suffers of some drawbacks, such as energy consumption, stability and locomotion speed, both if we are dealing with two, four, six or more legs. Hybrid systems were developed to exploit the terrain adaptability of legs in rough terrain and simpler control as well as high speed related to wheels.

Research efforts have been made toward hybrid mobile robots that can drive quickly and smoothly on flat terrain and can stably negotiate natural or artificial uneven terrain. Hybrid mobile robots can be classified into two main groups: in the first one legs (or sub-tracks) and wheels (tracks) work independently and articulated legs (sub-tracks) can provide traction for the obstacle-

climbing when wheels (tracks) cannot, or alternatively, wheels are purely passive and used to stabilize the system. This basic idea is used, for example, in [7–9] in which wheels are passive. Innovative prototypes of wheelchairs capable of climbing stairs have been proposed in [10,11]. A prototype of stair-climbing robot was proposed in [12]. The solution dealing with independent sub-track is adopted by QUINCE, which is a robot composed by four sets of wheels, each driving a track and powered by a total of six electric motors. It was developed by a research group from the Chiba Institute of Technology and Tohoku University to be used to collect rescue information in case of Chemical, Biological, Radiological and Nuclear disasters, like Fukushima nuclear plant.[13,14] Another relevant example is the Packbot,[15] which was developed and commercialized by IRobot, USA, for rescue and search application. It was successfully used into the damaged Fukushima nuclear plant after the 2011 Tōhoku earthquake and tsunami. It is composed by the mobile module and it can be accessorized by suitable modules for additional features, like chemical/radiation sensor payload, or combination of ground and aerial vehicles.[16]

The second type of hybrid locomotion involves wheels (or tracks) installed on each leg. In particular, the articulated leg is efficiently used to overcome slopes or obstacles that cannot be surpassed with tracks or wheels only. The design solution using wheels at leg tips was adopted, for example, in [17–20]. Another interesting design solution is the TITAN X [21], a novel leg-track hybrid mobile robot with a special leg-driving system

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installed on each leg to walk in leg form on rough terrain and move in tracked locomotion using track-belt on low-rough terrain. An example for vertical obstacle performance is Tri-Star IV,[22] which is a mobile robot designed to explore the moon surface in the near future. It is composed of two active arms and three active wheels and it can climb vertical obstacles through the two active arms.

The goal of this research is to develop a robot that can be used in industrial sites, especially in hazardous areas such as nuclear power plants. Other application areas are related to surveillance and rescue operations for military and civil tasks. Most of legged or wheeled robots, however, cannot be easily adapted to environments designed for human beings. If such devices are to be used at industrial sites, they will require special arrangements, such as ramps to allow them to move around. Since these places are originally designed for human beings, it is desirable to have robots capable of surpassing obstacles and adapt to the environment. Recent prototypes of hybrid robots using sub-tracks have been successfully used in such environments,[13,15] demonstrating the high adaptability of such systems. In this paper, we present a solution dealing with tracks and legs, indeed the contact with the obstacle during the climbing is obtained by the leg end-point, instead of a sub-track. The design concept of the new prototype of hybrid rover is shown in Figure 1 with low-cost easy-operation features. In order to obtain these characteristics for a walking machine the following requirements have been considered: compactness, light weight and reduced number of DOFs.

The leg structure shown in Figures 1 and 2 has 1-DOF and it is possible to combine more legs by sharing the same input crank. This solution allows the use of just one actuator for moving at same time four or more legs in order to design a robot capable of overcoming a large number of obstacle types.[23]

In particular, we have chosen to use a 1-DOF linkage to obtain a suitable motion of the leg tip to aid the rover

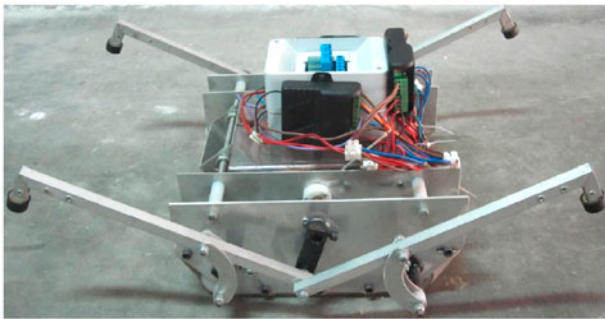


Figure 1. A built prototype of the proposed hybrid rover THROO.

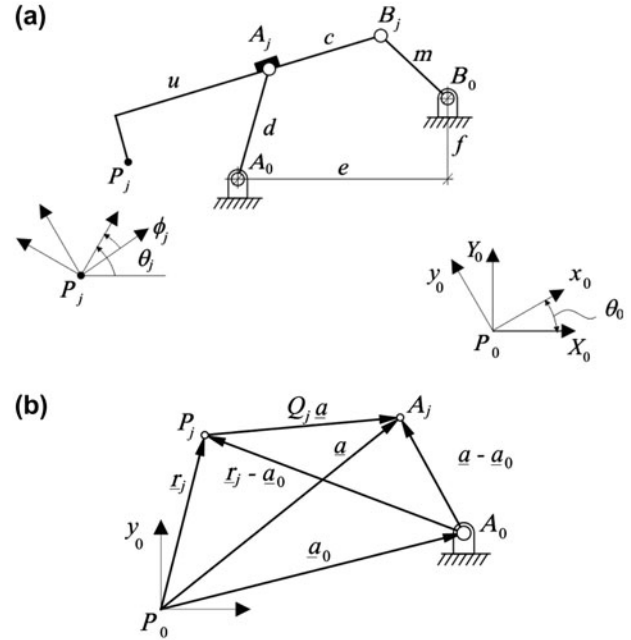


Figure 2. A kinematic scheme for the synthesis of a four-bar linkage: (a) in the j th pose for the rigid-body guidance with respect to the (P_0, ϕ_0) reference pose and (b) relation between points and vectors of the j th pose with those of the reference pose.

with rear and front legs to overcome obstacles. A synthesis procedure based on Burmester problem is used to obtain the leg mechanism according to design specification. A prototype has been designed and built and dynamic simulations and experimental tests have been performed to validate the operation of the proposed robot and its ability in overpassing obstacles.¹

According to the authors' knowledge, the most efficient solution for obstacle climbing and overpass is obtained by hybrid systems dealing with tracks and sub-tracks. Main difference of the proposed solution from the most efficient obstacle climbers, like for example QUINCE [13] or Packbot [15] is the leg design concept, since legs are in contact with the ground/obstacle through the leg endpoints, they have been designed to get a suitable motion of the leg to approach the obstacle and overpass it.

2. Kinematic analysis and synthesis of the leg mechanism

The hybrid rover has been designed with tracks and legs for an efficient locomotion with the ability to overcome obstacles. Among several solutions dealing with robotic legs, we have chosen to use a linkage having 1-DOF only for keeping the basic characteristics of the overall design, that are low cost and easy in use, in fact the

reduction of the number of DOFs leads to a complexity reduction in manufacturing and control. In the following, we use a synthesis procedure to obtain the leg dimensions and basics for Kinematics of the linkage.

2.1. Synthesis of the leg-mechanism

Planar mechanisms are in general used in industrial environment for automatic machinery in order to give prescribed laws of motion. They can be referred as function generators from the input to the output links, rigid-body guidance through the study of the rigid coupler motion and path generators by referring to the coupler curve, as extensively reported in [24–29]. Legs for mobile robots often have a design-containing mechanism and a class of slider-cranks can be also used for the purpose, according to the results proposed in [30]. A leg should generate an approximately straight-line trajectory of a foot point with respect to the body, as outlined in Refs. [31–33]. A number of design solutions for legs used a Chebyshev mechanism, as reported in [34–36], while the prototype proposed in [37] has 1-DOF leg consisting of single links having a circular trajectory. In this paper, we have chosen to synthesize the leg mechanism in order to get suitable size and characteristics of the leg end-point trajectory in terms of step height and step length. In this context, it is required to design a four-bar linkage that guides a rigid body attached to the coupler link through N prescribed poses. It is well known that a revolute-revolute dyad can be synthesized exactly for up to five prescribed poses.[38] The model deals with the Burmester problem, which aims at finding the geometric parameters of a four-bar linkage required for a prescribed set of finitely separated poses. In particular, the problem can be formulated as locating the positions of the joint centres A , A_0 , B and B_0 , in such a way that five given poses can be visited exactly.

The set of prescribed poses can be given with respect to a fixed frame, say OX_0Y_0 in Figure 2(a)), in the form (r_j, θ_j) , being r_j the position vector of a coupler point P at the j th pose, while θ_j defines the j th orientation of the guided body. If relative angles θ_j are expressed as $\phi_j = \theta_j - \theta_0$, and the origin of the fixed frame is chosen with O coincident with P_0 (first position of the coupler point P), then the set of N given poses reduce to (r_j, θ_j) . For four poses the problem can be formulated as a system of four algebraic equations in four unknowns, which are the coordinates of the joint centres A , A_0 , B and B_0 , as explained in [38]. Let us consider the scheme in Figure 2(b)).

The condition that the distance between points A_0 and A remains constant can be expressed as

$$\| (r_j - a_0) + Q_j a \| = \| a - a_0 \| \quad j = 1, \dots, N \quad (1)$$

in which a_0 and a represent the position vectors of points A_0 and A , respectively, and matrix Q_j can be expressed as

$$Q_j = \begin{bmatrix} \cos \phi_j & -\sin \phi_j \\ \sin \phi_j & \cos \phi_j \end{bmatrix} \quad j = 1, \dots, N \quad (2)$$

After squaring and expanding Equation (1) and further algebraic manipulation, the following synthesis equations can be obtained in the form [36]

$$f_j = a_0^T (1 - Q_j) a + r_j^T Q_j a - r_j^T a_0 + \frac{r_j^T r_j}{2} \quad j = 1, \dots, N \quad (3)$$

where $\mathbf{1}$ is the $[2 \times 2]$ identity matrix. Several authors have addressed the Burmester problem, it can be solved by intersecting two curves representing the loci of the centre points for two four-pose subsets out of the given five-pose set, as reported in [26,38,39], by applying complex numbers [40] and using kinematic mapping.[41] In the following, we have used the method proposed in [42] based on dyalitic elimination method, which leads to a numerically robust algorithm that can be successfully used for the case under study. In particular, joint centres can be found through the intersections of the four possible contours of the four-pose problem. Thus, the system of Equations in (3) can be rewritten in the form of dyalitic elimination of a being the vector of unknowns $q = [a^T \ 1]^T$.

$$Aq = 0 \quad j = 1, \dots, N \quad (4)$$

Therefore, matrix A admits solutions different from the trivial solution if A is non-singular. Thus, we compute the determinants of the four 3×3 sub-matrices. The four determinants are bivariate polynomials in the two unknown components of vector a_0 . The intersection of the four curves gives all possible real solutions of the problem. In a similar procedure vector a can be computed. It is shown in [42] that the four equations of any of the two sets reduce to a quartic polynomial.

According to the proposed algorithm, a four-bar linkage can be designed by considering the following design specifications: the point P is required to have the support phase of length b , with height h to overpass obstacles. Parameters b and h depend in general by the given task and overall dimension of the robot. Specifically, point P trajectory should have a suitable size in terms of high comparability with the body robot size for the flight phase, and length at least equal to the half length of the track size for an efficient use during the support phase. In addition, the trajectory should be almost straight line during support phase. Therefore, according to the chosen overall size of the robot, we have distributed the five poses according to these requirements: h is chosen equal to 160 and b has been chosen equal to 70 (both

expressed in mm). The first three poses have been chosen to lie on the support phase trajectory, one pose is chosen in the rise of the flight phase and one pose is chosen symmetrically on top of the rise.

According to the formulation, the curves leading to the A , A_0 , B and B_0 points being the intersection of the curves are shown in the plot of Figure 3. In particular, the four black plus four grey (eight) curves in Figure 3 represent the four determinants of the sub-matrices obtained from A eliminating dyalitically $\mathbf{a}$ or $\mathbf{a}_0$ vectors, respectively, as explained in the procedure (1)–(4).

2.2. Kinematics of the leg mechanism

The equation of a coupler point curve for any four-bar linkage may be obtained by analytic geometry being the loci of any point P that belongs to a segment for which points A and B are constrained to lie on two circles. According to the kinematic scheme in Figure 2(a), the position of point P , with coordinates $(x$ and $y)$ can be described with respect to OX_0Y_0 fixed frame, as a function of kinematic parameters of the mechanism A_0AB_0BP in the form of letters for table footnotes. In particular links A_0A (and B_0B) can be evaluated with respect to the OX_0Y_0 frame as given in (1), which can be rewritten by considering the fixed frame attached to point A_0 as

$$\begin{aligned} A_0A &= \begin{bmatrix} x + u \cos \theta - v \sin \theta \\ y + u \sin \theta + v \cos \theta \end{bmatrix} \\ B_0B &= \begin{bmatrix} x - p + (u + c) \cos \theta - v \sin \theta \\ y - q + (u + c) \sin \theta + v \cos \theta \end{bmatrix} \end{aligned} \quad (5)$$

Furthermore, by considering

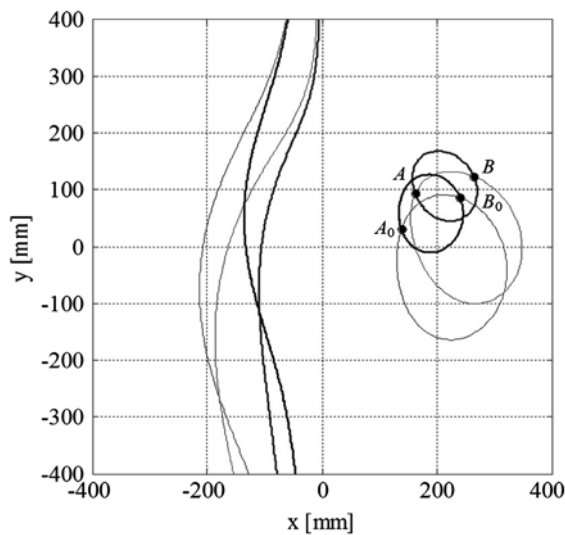


Figure 3. The eight curves giving the x and y coordinates of A , A_0 , B and B_0 points.

$$\|A_0A\|^2 = d^2 \quad \|B_0B\|^2 = m^2 \quad (6)$$

and substituting Equation (5) in Equation (6), one can obtain two equations, which are functions of design parameters and x , y and θ . The desired implicit function can be obtained by eliminating θ , and further algebraic manipulations. According to the reported formulation, the trajectory is shown in Figure 4 for the synthesized four-bar linkage of the b and h parameters. Coordinates of the five poses in Figure 4 are given in Table 1.

Design parameters of the synthesized mechanism obtained in the previous Section are reported in Table 2.

Results of the above mentioned study have been implemented in the robot body for its mechanical design, Figure 5 shows a planar scheme of the robot for which the front and rear legs are designed according to the dimensions in Table 2 and actuated by a single motor installed at point M chosen as coincident with B_0 . The common crank for the rear and front legs is depicted in black in Figure 5 and it is fixed to the chassis at point M .

3. Dynamic simulation

A suitable model of the hybrid rover has been developed in SolidWorks environment in order to test its operation, as it is shown in Figure 6. In particular, several paths with obstacles have been tested and simulated, by considering either regularly shaped obstacles or irregular terrain with obstacles. Figure 7 shows the snapshots of a reported dynamic simulation for the robot movement during its motion in overpassing a step obstacle of 100 mm, which represents the 125% of the robot track high.

It is worth noting that Figure 5 shows a lateral view (in which just one rear leg and one front leg are shown) but the hybrid rover in Figure 6 has been modelled and simulated in 3D to test the lateral stability also. In particular, to analyse the robot robustness with respect to possible overthrowing, several dynamic simulations have been carried out for several types of obstacles with different heights and sizes. Each simulation has rerun several times to demonstrate the engineering feasibility of the hybrid robot in overpass obstacles, the outcome of the experiments is: the robot can climb up to an obstacle that is 125% of the track high, the maximum tilting angle is 43° (for the reported simulation) and its maximum value before overturning is about 60° . It has been verified that the legs can be used to stabilize the hybrid robot during the motion and prevent longitudinal and lateral overthrow in climbing an obstacle.

The operation of the robot can be summarized as the following: (1) rover-like motion on flat or smooth terrain (Figure 7(d)), which is the most efficient type of locomotion under this condition and (2) walking-like motion when approaching and overcoming an obstacle (Figure 7(a)–(c)). In particular, the robot has been

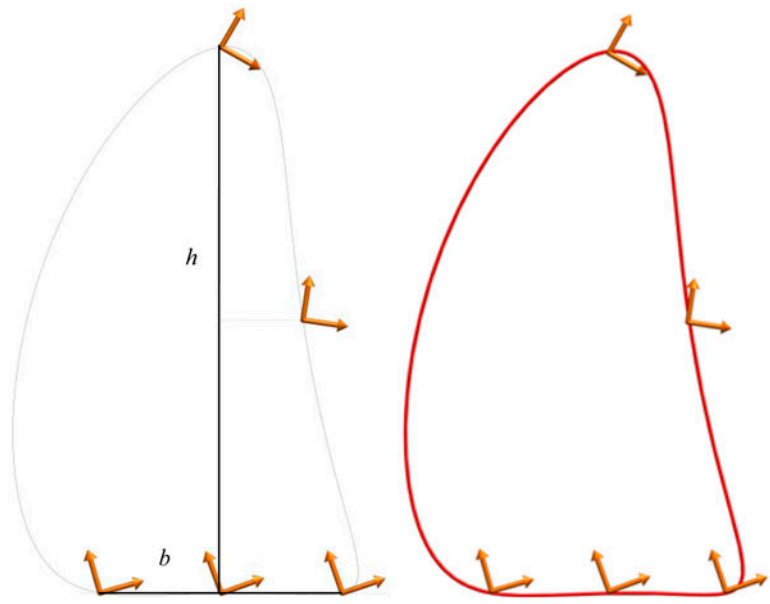


Figure 4. Trajectory of the synthesized four-bar linkage according to the given five poses with b and h parameters.

Table 1. Five chosen poses for the synthesis of the leg mechanism.

	x [mm]	y [mm]	ϕ [°]
Point n.1	$x = -240.0$	$y = 30.0$	18
Point n.2	$x = -275.0$	$y = 30.0$	20
Point n.3	$x = -310.0$	$y = 30.0$	16
Point n.4	$x = -275.0$	$y = 190.0$	330
Point n.5	$x = -250.0$	$y = 110.0$	350

provided by rear and front legs to stabilize it during the obstacle crossing (Figure 7(b), and (c)). Moreover, the rear leg is used as propulsive element in overpassing the obstacle. The main goal to pursue is, therefore, to use the front legs to approach the obstacle and rear legs to help the robot to overpass the obstacle.

Numerical results of the dynamic simulation have been reported in Figure 8, namely the rover CM position (Figure 8(a)) and CM velocity (Figure 8(b)).

Table 2. Kinematic parameters of the leg in Figures 2 (a) and 5.

Parameter	d	m	c	u	v	e	f
Dimension [mm]	66.0	45.2	105.8	166.2	63.5	104.2	40.0

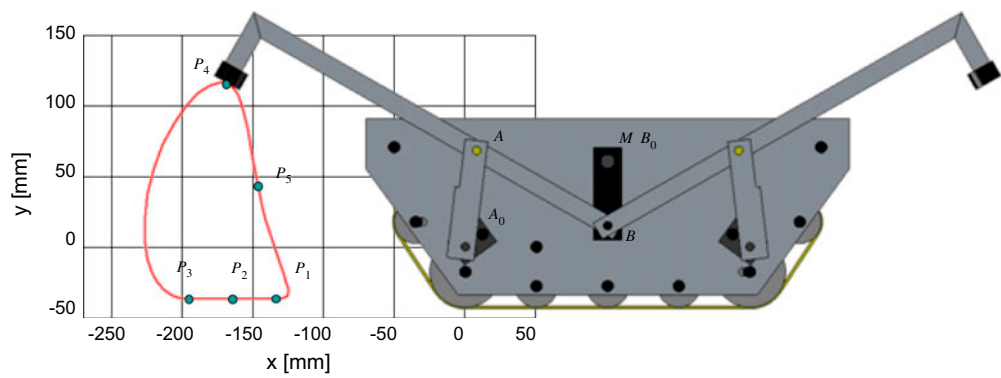


Figure 5. A scheme for the hybrid robot with coupler curve traced by point P .

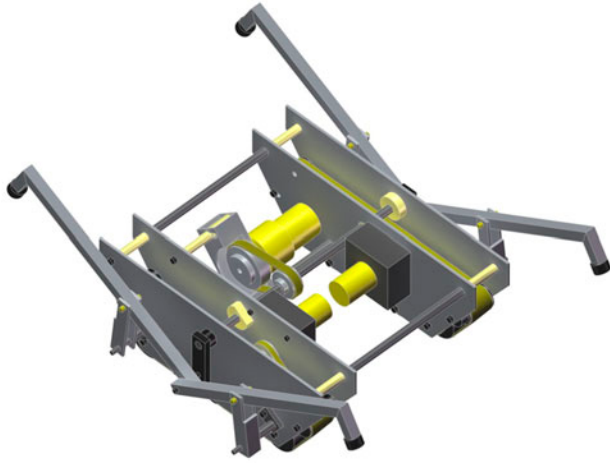


Figure 6. Mechanical design of the prototype of the hybrid rover.

Furthermore, the dynamic simulation allows to evaluate the torques required to give suitable power to the legs and track. By observing the simulation results on a flat surface the torque required for the legs is less than 0.3 Nm and the one needed for the track is less than 0.1 Nm. When the hybrid rover is overpassing the obstacle as shown in Figure 7, it can be seen that the maximum torque required for the legs is 4 Nm and it is less than 1 Nm for the track. This result was obtained thanks to the synthesized mechanism, in particular for the choice of y -coordinate for points P_1 to P_3 .

A good standard evaluation method for certifying locomotion characteristics of mobile robots is the step field pallet.[43] They are repeatable surface topologies to test the mobility of a robot. We have used a surface represented in Figure 9, when the dimension of a cube is $50 \times 50 \times 50$ mm. Among several stepfield pallets described in [43] that can be generated, we have chosen a regular topography in Figure 9 satisfying the standard stepfield's regulation in order to simulate an indoor environment. The field size used here was set as the standard step field's regulation. Figures 10 and 11 shows results of a dynamic simulation for the hybrid robot during a motion on the surface in Figure 9.

Nevertheless, this experiment can be considered as initial investigation can be further completed by testing the hybrid system with other topographies as described in [43].

4. A prototype of the hybrid rover

According to the results obtained from the dynamic simulation, a prototype of the hybrid rover has been designed and built considering the following constraints: minimum number of DOFs; low-cost mechanical design and modularity of the legs. According to those basic requirements a leg has been designed by considering robustness, reconfigurability, light weight and reduced number of DOFs. Currently, the robot is remotely operated, further developments will concern the implementation of control

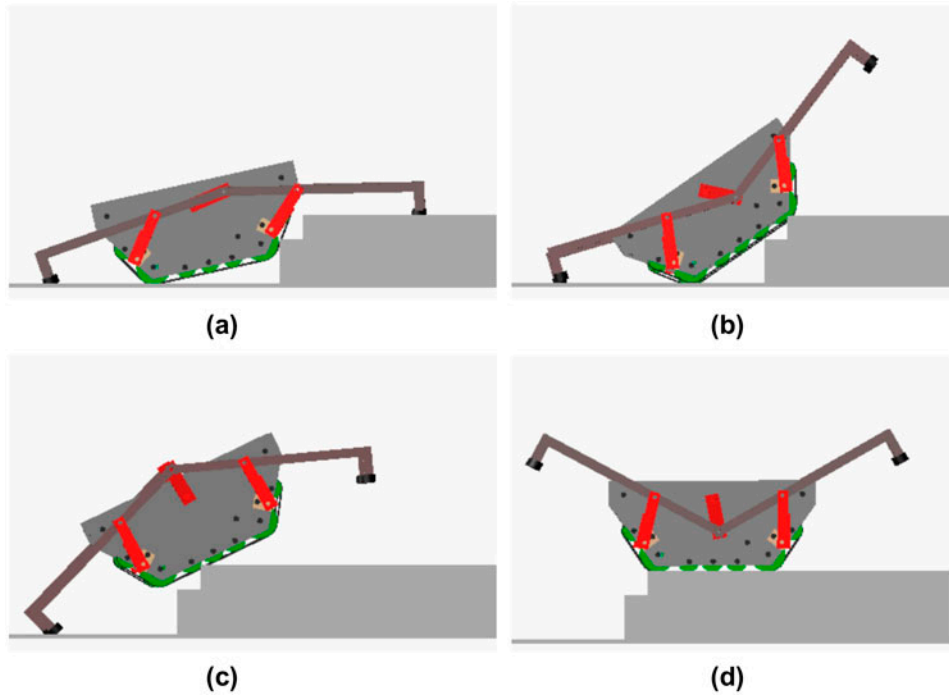


Figure 7. Dynamic simulation snapshots for the robot movement to overpass an obstacle of 100 mm: (a) 18 s; (b) 22 s; (c) 28 s and (d) 30 s.

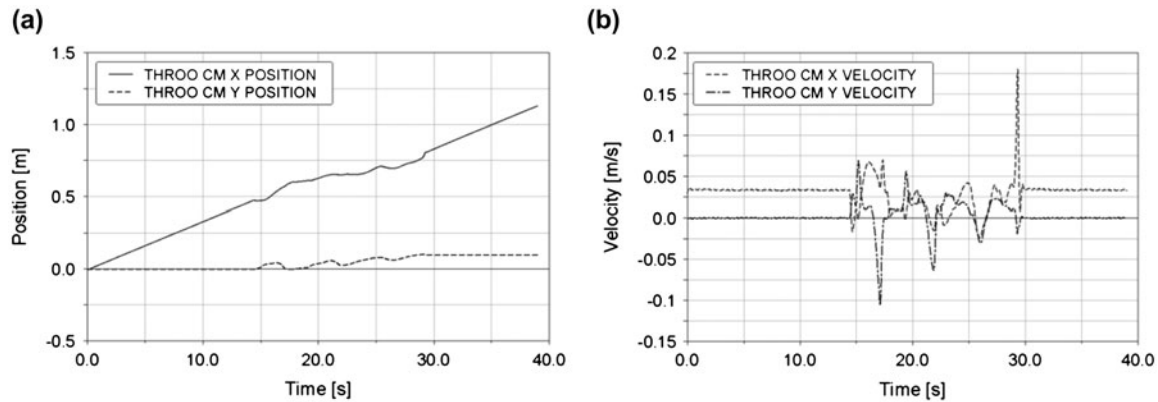


Figure 8. Results of the dynamic simulation in Figure 7 for the robot movement to overpass an obstacle: (a) CM position and (b) CM velocity.

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

Figure 9. Field designed for a test on the hybrid rover.

strategies for an efficient locomotion and to decide either overpass or circumnavigate an obstacle, as those reported in [44,45], which consist of novel control strategies designed for systems with large payload variations.

The proposed mechanical design offers interesting features to give the robot good performances. In particular, main characteristics are reported in Table 3, as regarding to the mechanical structure, mobility and additional features of the THROO robot and for two of most representatives and efficient prototypes available on the market that are Quince robot [46] and Packbot 510 robot.[47] In particular, they have in common the main structure to give mobility that is obtained by track, but then THROO has rear and front legs, Quince has rear and front sub-tracks and Packbot has front sub-tracks. Maximum obstacle to overpass depends on the main dimension of the robot and its height, the maximum

velocity and payload capacity depend on the actuation system also. In particular, both Quince and Packbot 510 show excellent features. Maximum inclination before overthrowing is comparable of the three robots.

Commercial solutions of Quice and Packbot 510 robots are equipped by several sensors. Quince has a 3-axis gravity sensor, gyroscope and PSD sensor, Packbot 510 is submergible in 3 feet of water, it has GPS compass, accelerometers and inclinometer. THROO robot has proximity sensors and encoders as this is a laboratory prototype. The three robots have hand-control.

4.1. Mechanical design

Figures 1 and 6 show a mechanical design of the robot with 3-DOFs, two of them are used for the track giving the ability of straight motion and turning ability. An additional motor is used to actuate the four legs, each couple of rear and front legs share the same crack and work providing a mirror trajectory of the leg end-point. A common shaft is used for left and right cranks. Belt transmission is used to drive the motion from motors to tracks and to the input cranks of the legs. Main characteristics are given in Table 3.

Motors have been installed to the robot chassis in order to distribute uniformly their masses. According to the proposed mechanical design and suitable mass distribution, the centre of mass of the robot lies in the longitudinal plane of robot body, while its y -coordinate varies according to the legs' configuration. Its average relative y -coordinate is 55 mm, with respect to the leg reference frame shown in Figure 5.

It is important to point out that as a design requirement the robot should work in both locomotion directions having the same ability of overpass obstacles. This is to increase the robot's operation flexibility.

According to initial design idea and simulation results, during the rover-like motion the legs should be inactive and in the walking-like motion either legs or

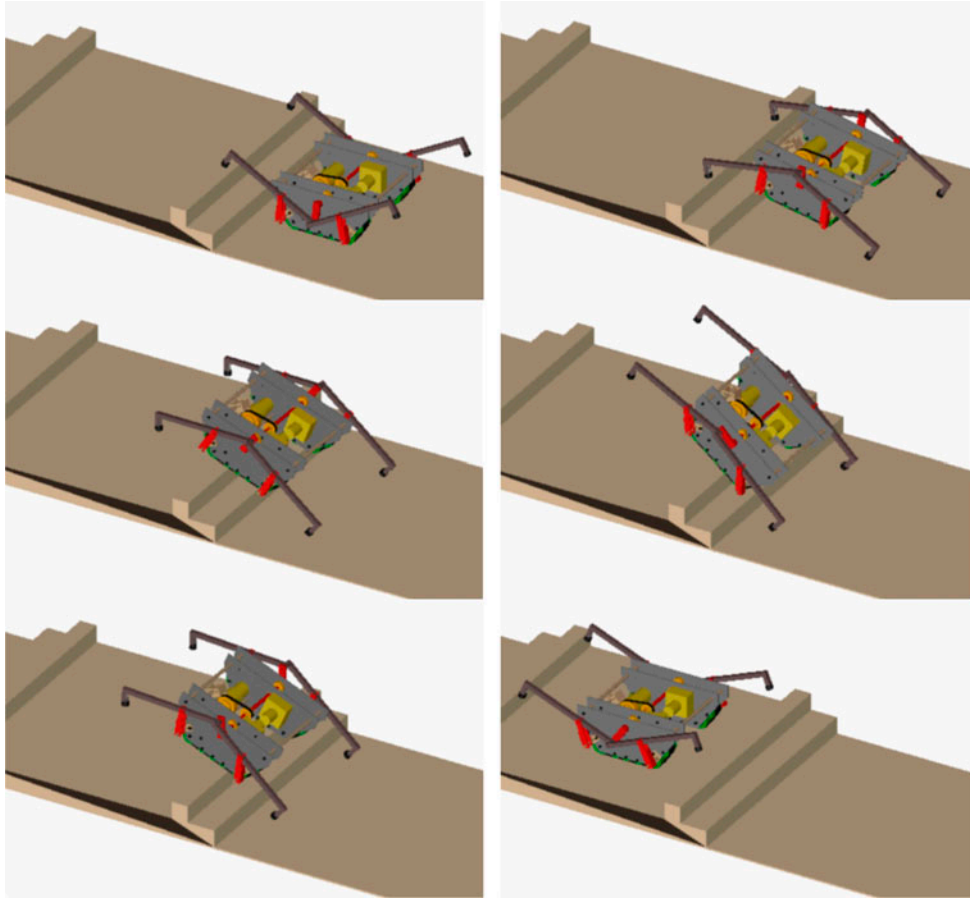


Figure 10. Dynamic simulation snapshots for the robot to overpass the field in Figure 9.

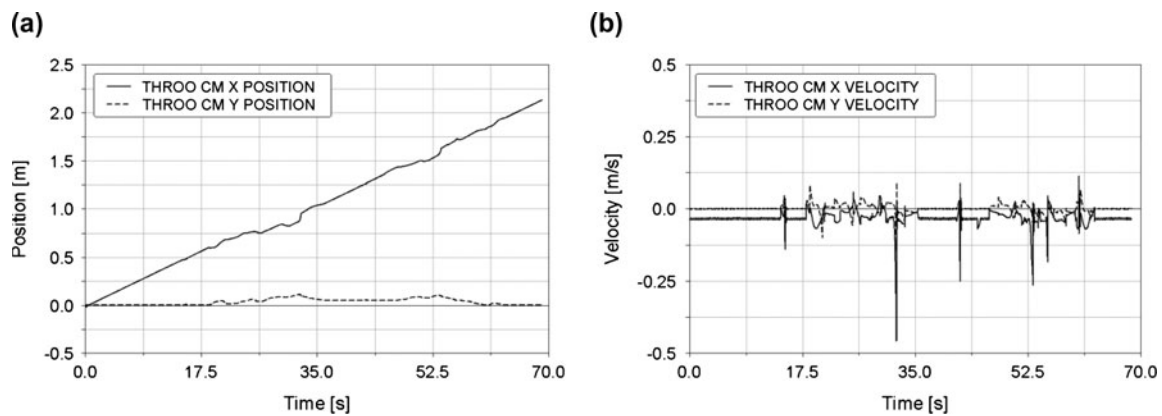


Figure 11. Results of the dynamic simulation in Figure 10 for the robot movement to overpass the field in Figure 9: (a) CM position and (b) CM velocity.

tracks should be active. It worth to note that even if the legs and tracks operate simultaneously on flat surfaces, the leg end-points do not interact with the terrain, as it is shown in the scheme of Figure 5 and in the simulation results, which are reported in Figures 7 and 10.

4.2. Experimental tests

A prototype of the THROO hybrid rover has been built according to the results of the dynamic simulation proposed in the previous section. Its 3-DOFs are currently remote controlled and future development of

Table 3. Comparison of hybrid robots' specifications.

		Quince	510 Packbot	THROO
Robot size (in mm) and weight	Length	1110	686 flippers stowed 889 flippers extended	Body: 300; body legs extended: 550
	Width	420	406 no flippers 521 with flippers	400
	Height	420/210 (antenna folded)	178 with no payload or manipulator	Body: 140; body legs extended: 200
	Weight (DOFs)	27 kg 2 (track) 4 (sub-track)	10.89 kg without batteries 2 (track) 2 (sub-track)	4.5 kg without batteries 2 (track) 1 (legs)
	Material	Carbon, Aluminium alloy	Not available	Aluminium alloy, steel alloy
Travelling performances	Max speed	1.1 m/s	Up to 5.8 mph	Up to 0.5 m/s
	Max step	340 mm	Not available	100 mm
	Max inclin	60°	60°	60°
	Autonomy	3 h	more than 4 h	2 h
Main track (size in mm)	Length	710	686	260
	Width	165	406	280
	Motor	MAXON EC-PowerMax30 (200 W/36 V)	Not available	DC motor 24 V, 5 Nm, 3.9 W × 2
	Material	Rubber, Aluminium alloy	Not available	Rubber, Aluminium alloy
Sub-track or Leg (size in mm)	Length	345	413	272
	Width	28	55	10
	Motor	MAXON EC 22 (50 W/32 V)	Not available	DC motor 24 V, 12 Nm, 24 W
	Material	Rubber, Aluminium alloy	Not available	Rubber, Aluminium alloy

the robot is its sensorization in order to automatically activate the legs just in proximity of obstacles. The prototype size is $550 \times 400 \times 200$ mm, all mechanical components are made of aluminium and actuation and transmission systems are of commercial type. The overall

mass of the prototype is 4.5 kg (batteries not included). It is worth to note that its dimensions vary according to the legs' configuration, here maximum values are given. The actuation system is composed by 2 DC motors for the track with following features: power supply of 24 V,

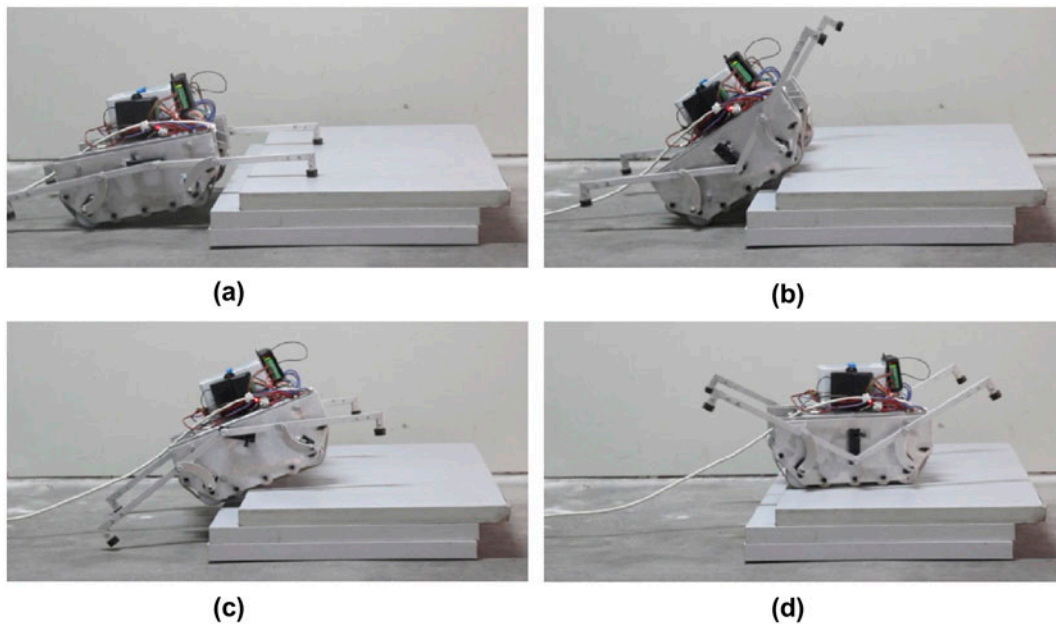


Figure 12. Experimental test of the THROO robot for overpass an obstacle of 100 mm, which is 125% of the track high. Note: You can watch these videos at: <http://www.youtube.com/user/RoboticaLab>.

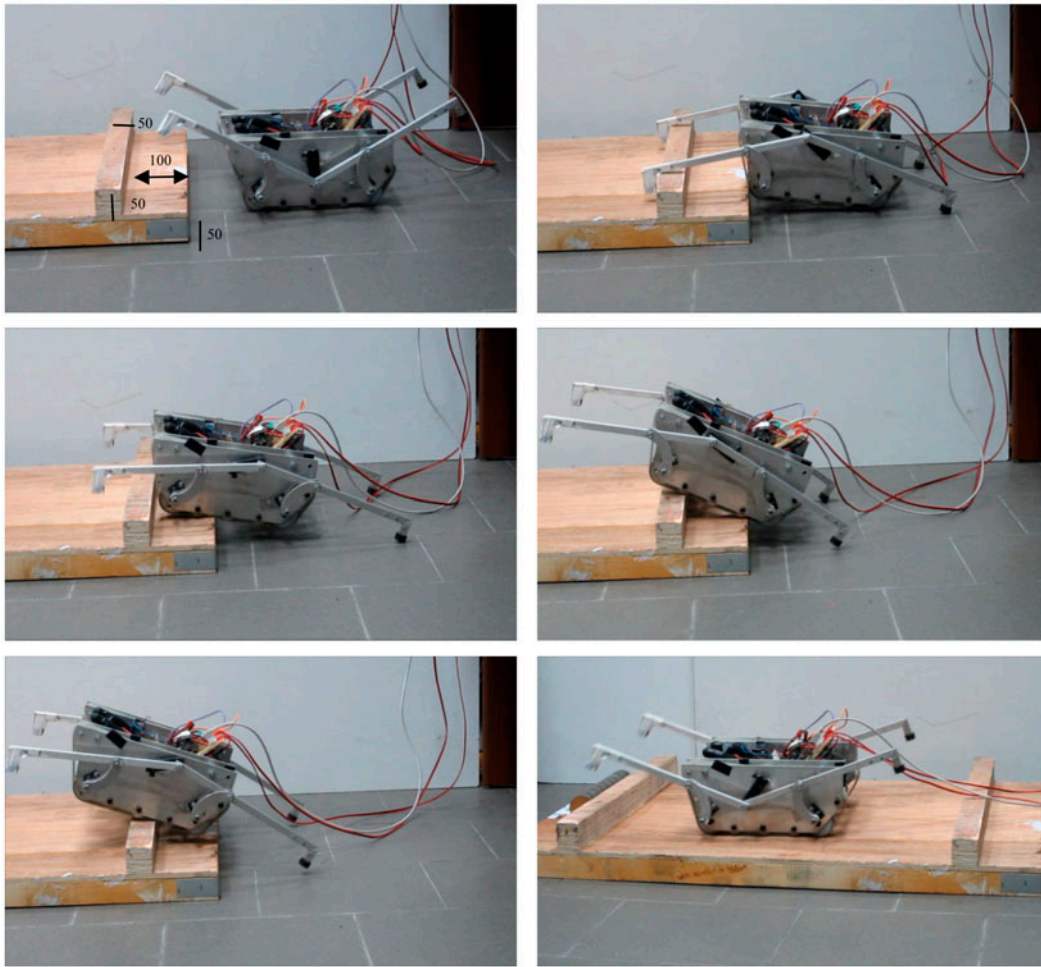


Figure 13. Experimental test of the THROO robot for overpass the field in Figure 9.

max torque of 5 Nm and nominal power of 3.9 W. The motor used to actuate the legs has a power supply of 24 V, max torque of 12 Nm and nominal power of 24 W.

The THROO hybrid robot has been tested in several operating conditions. A comparison between numerical and experimental tests is performed and it gives a good match of expected results. Several experimental tests have been carried out in indoor and outdoor environments. In particular, it has been experimentally tested for the ability of the hybrid robot in overpass obstacles. Figure 12 shows snapshots of the hybrid robot to overpass the same obstacle in Figure 7, which is greater than the track size.

Figure 13 shows experimental tests obtained by using a step field pallet represented in Figure 9. A comparison between numerical and experimental results of the hybrid robot for both proposed tests shows a good match providing a useful tool for the simulation and off-line programming and control of the robot.

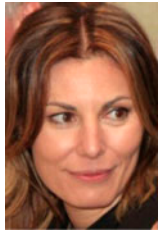
5. Conclusion

The design and construction of the THROO hybrid rover have been presented and discussed in this paper. A synthesis algorithm has been used to design the leg mechanism that is based on a four-bar linkage. Front and rear legs have been designed to work efficiently to overpass obstacles, both to pull the obstacle with front legs and also to use rear legs as propulsion system to push terrain. Main feature, which can be considered as a novel characteristic for the proposed hybrid rover, is its capability of overpass obstacles, which are greater than the dimension of its tracks. Future development will concern a suitable sensorization of the robot to provide the ability of automatically switch from rover-like to walking-like modes and vice versa in the proximity of an obstacle. Further developments that concern robot control strategies are to be implemented in order to decide to overpass or circumnavigate an obstacle.

Note

1. You can watch these videos at: <http://www.youtube.com/user/RoboticalLab>.

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